NFL Home-Win Prediction Models

Training: 2022-2024 regular season, weeks 6-18 (n = 576 games after removing 1 tie and 2 rows with missing rest data). Hold-out: 2025 regular season, weeks 6-18 (194 games). Outcome Y = 1 if the home team won, 0 if the away team won. Predictions at the 0.5 threshold.

Notation: **wr_diff** = home_last5_win_rate – away_last5_win_rate. **rest_less** / **rest_more** are indicator dummies from the ordinal rest_period_diff (baseline = rest_same).

Model 1 — OLS, wr_diff only (baseline)

Formula:

```
Y■ = 0.5735 + 0.4156 · wr_diff
```

Predict home win if Y■ ≥ 0.5.

Coefficients:

term	estimate	SE	stat	p-value
intercept	+0.5735	0.0197	t = +29.13	~1e-186
wr_diff	+0.4156	0.0540	t = +7.69	1.5e-14

R^2 = 0.093 · Pearson r = +0.306 · residual SD = 0.472

Model 2 — Logistic, wr_diff only (baseline)

Formula:

$$P(\text{home win}) = \frac{1}{1 + e^{-(0.3271 + 1.8526 \cdot \text{wr_diff})}}$$

Predict home win if P ≥ 0.5.

Coefficients:

term	estimate	SE	stat	p-value
intercept	+0.3271	0.0890	z = +3.67	2.4e-4
wr_diff	+1.8526	0.2635	z = +7.03	2.0e-12

Odds ratio per +1.0 in wr_diff = 6.38; per +0.20 = 1.45.
McFadden pseudo-R^2 = 0.071 · LR $\chi^2(1)$ = 56.2 ($p \approx 6e\text{-}14$) · Brier = 0.222 · in-sample accuracy = 64.4%

Predicted P(home win) across wr_diff (logistic):

wr_diff	-0.80	-0.40	-0.20	0.00	+0.20	+0.40	+0.80
P(home win)	24.0%	39.8%	48.9%	58.1%	66.8%	74.4%	85.9%

2025 hold-out accuracy (baseline models)

Both baseline models classify identically at the 0.5 threshold (0 disagreements): the OLS boundary sits at wr_diff = -0.177, the logistic boundary at -0.176.

window	model	correct	accuracy	calibration
weeks 6-18	OLS (wr_diff)	135 / 194	69.6%	Brier = 0.2087
weeks 6-18	Logistic (wr_diff)	135 / 194	69.6%	log-loss = 0.6043
weeks 10-18	OLS (wr_diff)	99 / 137	72.3%	Brier = 0.2015
weeks 10-18	Logistic (wr_diff)	99 / 137	72.3%	log-loss = 0.5880
weeks 6-18	always-home baseline	103 / 194	53.1%	
weeks 10-18	always-home baseline	73 / 137	53.3%	

Explored models with rest_period_diff (not adopted)

Adding the ordinal rest-period comparison as covariates. Baseline for the dummies is rest_same (home team's rest_period equal to away's).

Model 3 — OLS with rest_less + rest_more

```
Y■ = 0.5905 + 0.4227 · wr_diff − 0.1218 · rest_less + 0.0028 · rest_more
```

term	estimate	SE	stat	p-value
intercept	+0.5905	0.0231	t = +25.51	~1e-143
wr_diff	+0.4227	0.0541	t = +7.82	5.4e-15
rest_less	-0.1218	0.0571	t = -2.13	0.033
rest_more	+0.0028	0.0582	t = +0.05	0.96

$R^2 = 0.101$

Model 4 — Logistic with rest_less + rest_more

$$P(\text{home win}) = \frac{1}{1 + e^{-(0.4044 + 1.9022 \cdot \text{wr_diff} - 0.5488 \cdot \text{rest_less} + 0.0232 \cdot \text{rest_more})}}$$

term	estimate	SE	stat	p-value
intercept	+0.4044	0.1056	z = +3.83	1.3e-4
wr_diff	+1.9022	0.2665	z = +7.14	9.4e-13
rest_less	-0.5488	0.2575	z = -2.13	0.033
rest_more	+0.0232	0.2657	z = +0.09	0.93

McFadden $R^2 = 0.077$ · LR $\chi^2(3) = 61.0$ ($p \approx 4e\text{-}13$)

Model 5 — OLS with rest_less only (rest_more dropped)

```
Y■ = 0.5910 + 0.4228 · wr_diff − 0.1222 · rest_less
```

term	estimate	SE	stat	p-value
intercept	+0.5910	0.0212	t = +27.87	~1e-170
wr_diff	+0.4228	0.0540	t = +7.83	4.7e-15
rest_less	-0.1222	0.0562	t = -2.17	0.030

$R^2 = 0.101$

Model 6 — Logistic with rest_less only (rest_more dropped)

$$P(\text{home win}) = \frac{1}{1 + e^{-(0.4081 + 1.9029 \cdot \text{wr_diff} - 0.5524 \cdot \text{rest_less})}}$$

term	estimate	SE	stat	p-value
intercept	+0.4081	0.0971	z = +4.20	2.7e-5
wr_diff	+1.9029	0.2663	z = +7.14	9.0e-13

rest_less	-0.5524	0.2541	z = -2.17	0.030

McFadden $R^2 = 0.077$ · OR(rest_less) = 0.58 — when the home team has strictly less rest than the visitor, home-win odds are multiplied by ~0.58.

2025 hold-out accuracy (with rest_less)

window	model	correct	accuracy	calibration
weeks 6-18	OLS + rest_less	127 / 194	65.5%	Brier = 0.2085
weeks 6-18	Logistic + rest_less	127 / 194	65.5%	log-loss = 0.6033
weeks 10-18	OLS + rest_less	90 / 137	65.7%	Brier = 0.2040
weeks 10-18	Logistic + rest_less	90 / 137	65.7%	log-loss = 0.5928

The rest_less coefficient is nominally significant on the training set ($p \approx 0.03$) and slightly improves probability calibration on 2025, but 0.5-threshold accuracy drops 4 pp (69.6→65.5) at weeks 6+ and 6.6 pp (72.3→65.7) at weeks 10+. Adopted baseline: Models 1 & 2.

Source data: dev_data/metrics/{2022..2025}_all_game_metrics.csv. Coefficient estimation: OLS via normal equations; logistic via Newton-Raphson to convergence (5 iterations). Standard errors are OLS/model-based; large-sample normal approximation used for p-values.